BIOL 343
Applied Bioinformatics

DEG analysis

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Counting converts mapped reads to expression values

Expression values will be used by statistical models to identify DEGs

- Recall the goal of our RNA-seq experiments…
  - Treatment vs Control
  - Mutant vs Wild type
  - ***Identify differentially expressed genes (DEGs)***
- DEGs will be identified using statistical tests comparing ***expression values*** of transcripts/genes
- Expression values will be calculated based on the number of reads that ***align/map*** to a given genomic locus
- ***So you have DEGs…now what?***

DEGs are analyzed and evaluated by content experts

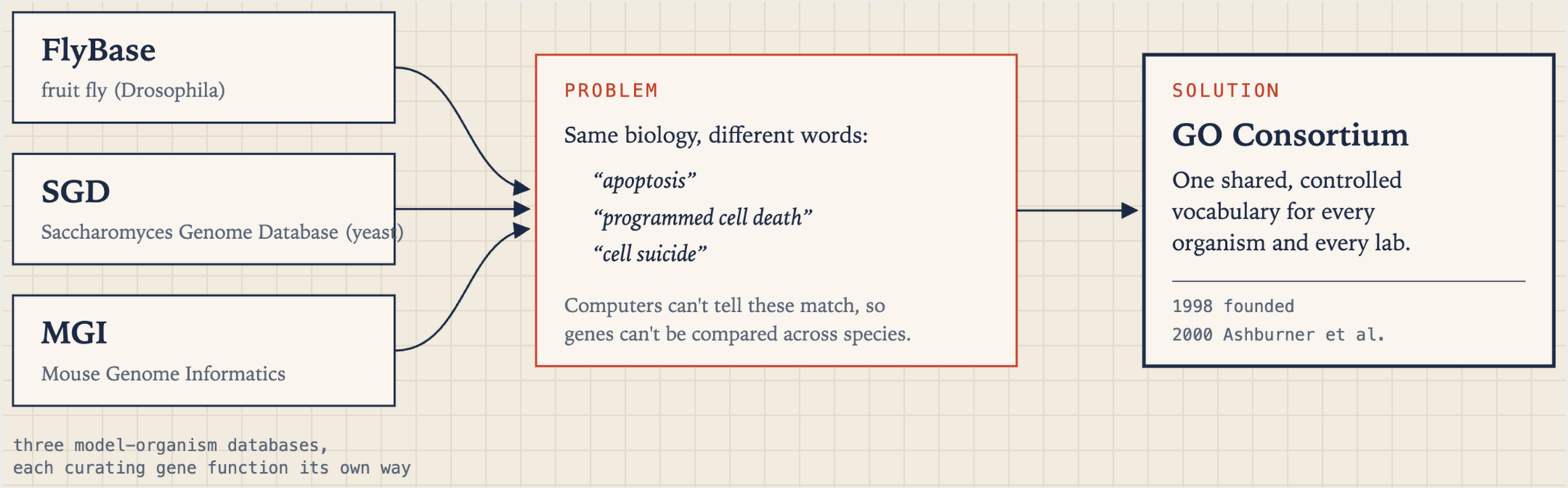
Sometimes they are a bioinformatician, sometimes not

- DEGs are list of genes that ***may*** be important in the response to a perturbation (treatment, mutation, etc.)
- Just because they are differentially expressed does not mean they are biologically meaningful in that context
- Lists of DEGs are provided to the subject matter experts - the people who designed the experiment
- Subject matter experts explore DEGs to build new hypotheses
- DEGs hypothesized to be responsive to the perturbation need to be followed up with new experiments…otherwise it is just speculation
- There are a few ways that almost all RNA-seq analyses will begin to explore DEGs to look for biological relevance…

Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

- In the 1990s, researchers needed a way to quickly link DNA sequence to the potential activity and function of the gene product
- In 1998, three model organism genome databases (flies, yeast, mice) collaborated to start the Gene Ontology Consortium



Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

Molecular Function

What does the molecule do?

e.g. kinase activity, DNA binding

`root GO:0003674`

Biological Process

What larger program is it part of?

e.g. DNA repair, cell cycle

`root GO:0008150`

Cellular Component

Where in the cell does it act?

e.g. nucleus, mitochondrion

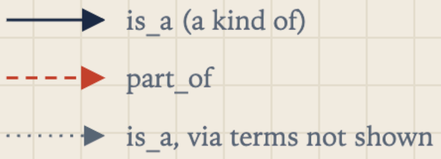
`root GO:0005575`

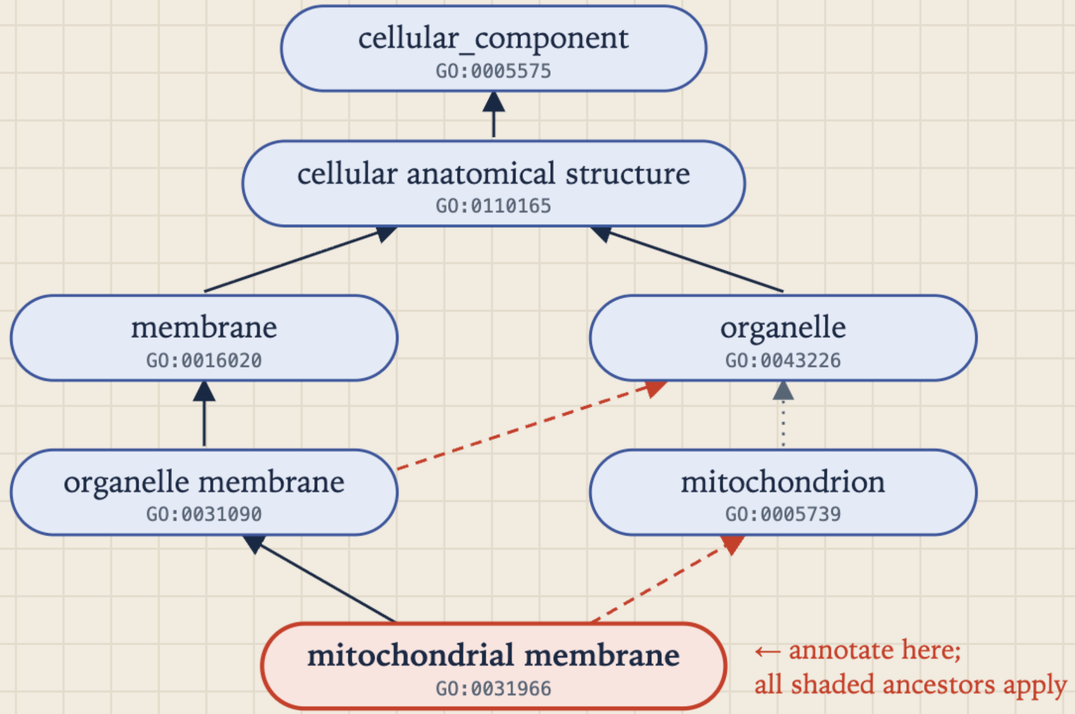
ZOOM: PART OF THE CC GRAPH

Terms run from general (top) to specific (bottom). A term can have more than one parent, so it's a graph, not a tree.

True path rule

Annotate a gene to one term and it's implicitly annotated to every ancestor, too.





Gene Ontology (GO) annotations

A vocabulary for describing what gene products do

GENE PRODUCT

TP53

human tumor suppressor protein

+

GO TERM

DNA-binding transcription factor activity

`GO:0003700 · Molecular Function`

+

EVIDENCE CODE · six categories, one example each

Experimental	**IDA**	direct assay (+ HTP variants)
Phylogenetic	**IBA**	biological aspect of ancestor
Computational	**ISS**	seq. or structural similarity
Author statement	**TAS**	traceable author statement
Curator statement	**IC**	inferred by curator
Electronic	**IEA**	not manually reviewed

+

REFERENCE

The paper or method behind the claim

↓ multiply by millions of annotations → enables

Cross-species comparison

same term means the same thing everywhere

Enrichment analysis

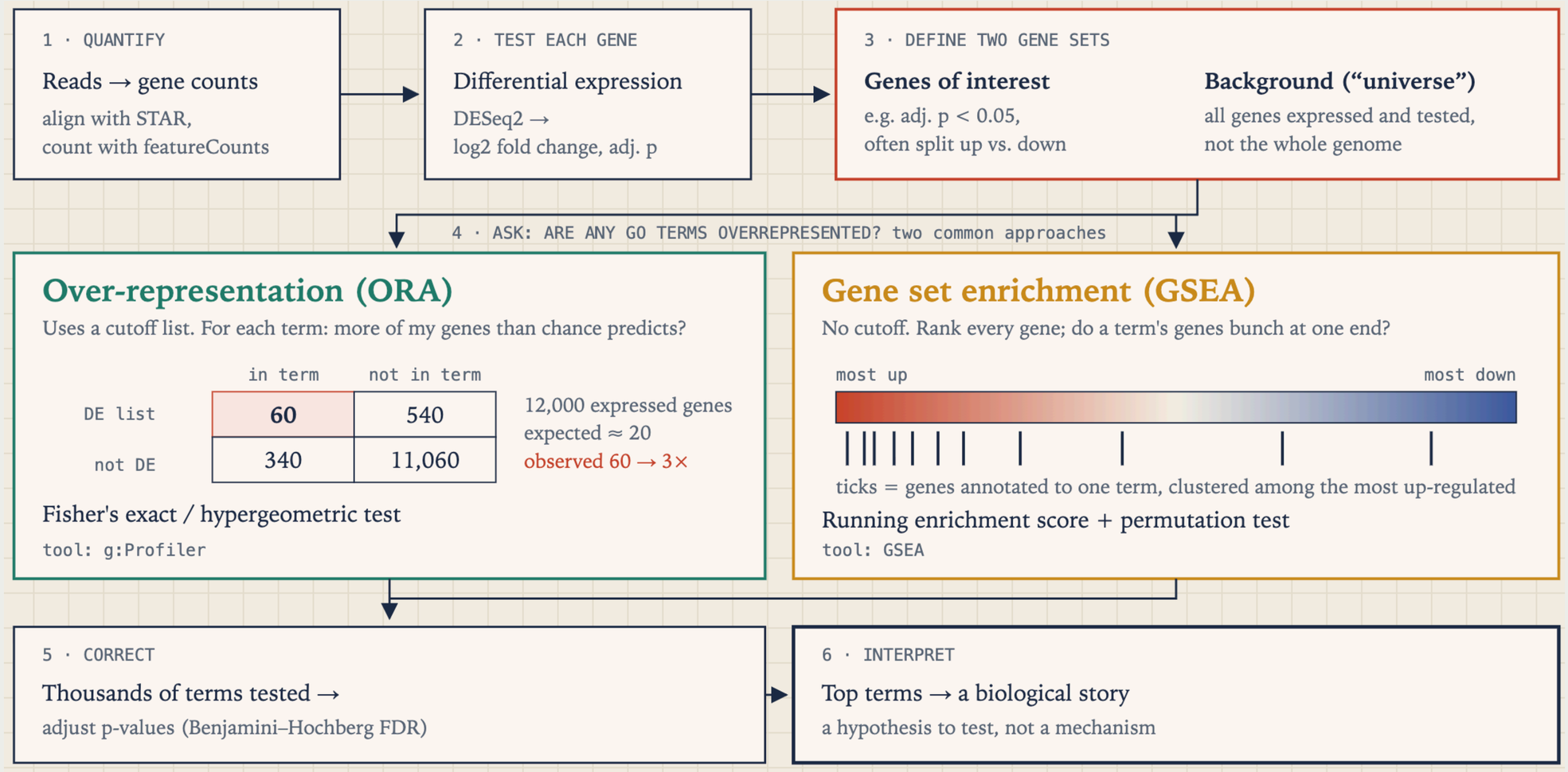
which functions are overrepresented in a list?

Weighing the evidence

experiment vs. inference, per claim

Over-representation and gene set enrichment

Using statistics to identify GO terms that are enriched/unenriched in lists of DEGS



https://zfin.org/ZDB-GENE-061103-301